

Bitan Majumder

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ABOUT

M.Sc. Computer Science student with research experience in Natural Language Processing and Machine Learning. Interested in applying NLP methods to problems in Computational Social Science. Previously I worked on multilingual and code-mixed text. Experienced in model evaluation, fine-tuning, and data curation for real-world language analysis tasks.

EDUCATION

Master of Science in Computer Science CGPA: 8.7/10 (up to 3rd sem) <i>Pondicherry Central University</i>	July 2024 – May 2026 <i>Kalapet, Pondicherry</i>
Bachelor of Science (Honours) in Computer Science CGPA: 8.8/10 <i>Ramakrishna Mission Vivekananda Centenary College</i>	Sept 2021 – May 2024 <i>Rahara, Kolkata</i>
WBBSE: 88% WBCHE: 89% <i>Ramakrishna Mission Vidyapith</i>	2013 – 2021 <i>Purulia, WB</i>

SKILLS

- Programming:** Python, C, Java, SQL
- Deep Learning:** PyTorch, TensorFlow, Keras
- Graph & Language Modeling:** NLP, Graph Neural Networks, PyTorch Geometric
- Web & Markup Technologies:** HTML, LaTeX
- Version Control:** Git, GitHub
- Languages:** Bengali (Native), English (Fluent), German (Elementary)

EXPERIENCE

Intern <i>Ashoka University</i>	Jan 2025 – Present <i>Remote</i>
- Conducted empirical research on sarcasm detection in Hinglish (Hindi-English) code-mixed text, comparing four large language models (Llama 3.1, Mistral, Gemma 3, Phi-4) with a fine-tuned DistilBERT model. - Curated a balanced and manually translated dataset, addressing data scarcity in low-resource, code-mixed language settings. - Implemented multiple fine-tuning strategies, including sarcasm-only training and sentiment-based transfer learning, to enhance model performance. - Achieved 84% classification accuracy using sequentially fine-tuned DistilBERT, significantly outperforming all LLM baselines. - Demonstrated that domain-adaptive fine-tuning significantly outperforms zero-shot LLM inference for code-mixed sarcasm detection.	
Intern <i>IEEE Computer Society Student Branch Chapter, IEEE Kharagpur Section</i>	June 2024 – Aug 2024 <i>Remote</i>
- Conducted in-depth research on time series forecasting, with a focus on evaluating Large Language Models (LLMs) as zero-shot predictors. - Reproduced and extended methodologies from the paper “Large Language Models Are Zero-Shot Time Series Forecasters” to design experimental forecasting pipelines. - Implemented forecasting workflows using Gemini and OpenAI APIs, adapting prompt-based inference for structured temporal data. - Engineered and preprocessed custom time series datasets in Python to systematically assess model performance. - Analyzed forecasting accuracy and limitations of foundation models in temporal prediction tasks, gaining practical insight into LLMs for non-text modalities.	

Research Intern*Indian Institute of Engineering Science and Technology (IIST)*

June 2023 – Dec 2023

Shibpur, India

- Designed and developed an automated exam answer script evaluation system using Natural Language Processing (NLP) and unsupervised machine learning techniques.
- Implemented text preprocessing, feature extraction, and similarity-based scoring methods to predict marks for descriptive answers.
- Conducted experiments to compare model-predicted scores with human-evaluated marks, achieving competitive and consistent performance.
- Gained hands-on experience in applying machine learning to real-world educational assessment problems, focusing on fairness and scalability.

Research Intern*West Bengal State University*

Aug 2022 – Feb 2023

Kolkata, India

- Contributed to the development of an Optical Character Recognition (OCR) system for the Bengali language, addressing challenges in low-resource scripts.
- Designed and implemented character segmentation pipelines using contour detection, smoothing, and noise removal techniques.
- Applied image processing algorithms to segment individual Bengali characters from scanned document images.
- Utilized OpenCV and Python to preprocess datasets and improve segmentation robustness.
- Gained strong foundations in digital image processing and practical experience in building OCR systems for Indic languages.

PROJECTS**Automated Exam Paper Evaluation (NLP, ML):**

- Designed an automated evaluation system for descriptive exam answers using NLP and unsupervised machine learning, reducing manual grading bias and effort.
- Implemented Doc2Vec-based document embeddings trained on subject-specific textbooks to capture semantic similarity between student answers and model answers.
- Applied DBSCAN clustering on answer embeddings to group similar responses and handle outliers, enabling scalable and fair grading.
- Computed cosine similarity-based scoring and mapped similarity scores to final marks, achieving close agreement with human evaluation (low MSE across questions).
- Conducted comparative experiments with BERT, SciBERT, Word2Vec, and Doc2Vec, justifying model selection based on performance and data constraints.

Code-mixed Sarcasm Detection in LLMs:

- Investigated sarcasm detection in multilingual and code-mixed (Hinglish) text, addressing challenges from informal language, grammar variation, and low-resource settings.
- Benchmarked four large language models (LLaMA 3.1, Mistral, Gemma 3, Phi-4) against a fine-tuned DistilBERT model for sarcasm classification.
- Constructed a balanced, manually translated Hinglish dataset to ensure reliable evaluation in a low-resource code-mixed scenario.
- Showed that domain-adaptive and sentiment-based fine-tuning significantly improves sarcasm detection over general LLM inference.

CERTIFICATIONS

- Deep Learning Specialization - Deeplearning.ai
- IELTS (Band: 8) - IDP

RELEVANT LINKS**LinkedIn:** : <https://www.linkedin.com/in/bitam-majumder-672801237>**Portfolio:** : <https://bitanm.github.io/>**GitHub:** : <https://github.com/bitamM>**Kaggle:** : <https://www.kaggle.com/bm2002>**HackerRank:** : <https://www.hackerrank.com/profile/bitamajumder201>